



The University of Texas at Austin
McCombs School of Business

AgroScope: Plant Seedling Classifier

Convolutional neural networks for automated field monitoring

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Introduction to Computer Vision · May 2024

Post Graduate Program in AI & ML: Business Applications

McCombs School of Business, The University of Texas at Austin

PROJECT

05

Introduction to Computer Vision

A convolutional neural network classifying 4,750 seedling images into 12 species; eight architectures compared, from a 71% baseline to a 92% regularized, class-weighted model.

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Executive summary

92%

test accuracy — final CNN, 475
held-out images

71%

baseline accuracy — three-layer
CNN, no augmentation

8

architectures trained, including two
transfer-learning attempts

12

species; 4,750 images at 128×128
→ 64×64

Key findings

Regularization and class weighting did the work. L2 on every layer after the first, batch normalization, dropout and balanced class weights lifted accuracy by 21 points over the baseline with the same input size.

Transfer learning failed outright. VGG16 and ResNet50 with frozen bases scored 14% — the majority class. At 64×64 the ImageNet features do not transfer without fine-tuning.

One confusion remains. Black-grass and Loose Silky-bent look alike; Black-grass recall is 0.50 while every other class is 0.83 or better.

Recommendations

01

Deploy in the field

Mobile app for real-time species identification on-site by farmers and botanists; integrate with existing farm-management systems.

02

Automate the data pipeline

Collect new images continuously and retrain, so the model adapts to new regions, seasons and species.

03

Train and support users

Sessions for end users on capability and limits; a feedback loop for misclassifications to feed retraining.

04

Invest in Black-grass data

More, higher-quality images of the two confused species in varied conditions; texture or spectral features; targeted augmentation such as color shift and perspective.

05

Keep tuning regularization

L2 was the decisive change. Continue to monitor regularization strength and learning-rate schedule as data grows.

06

Revisit transfer learning properly

Fine-tune rather than freeze, at native resolution; evaluate ResNet or hybrid VGG-ResNet features before ruling them out.

Business problem and solution approach

Problem

- Agriculture still relies on people to walk fields and tell seedlings from weeds. Identification is slow, labor-intensive and hard to scale.

Objective: a CNN that classifies a seedling image into one of twelve species accurately enough to automate monitoring.

Approach

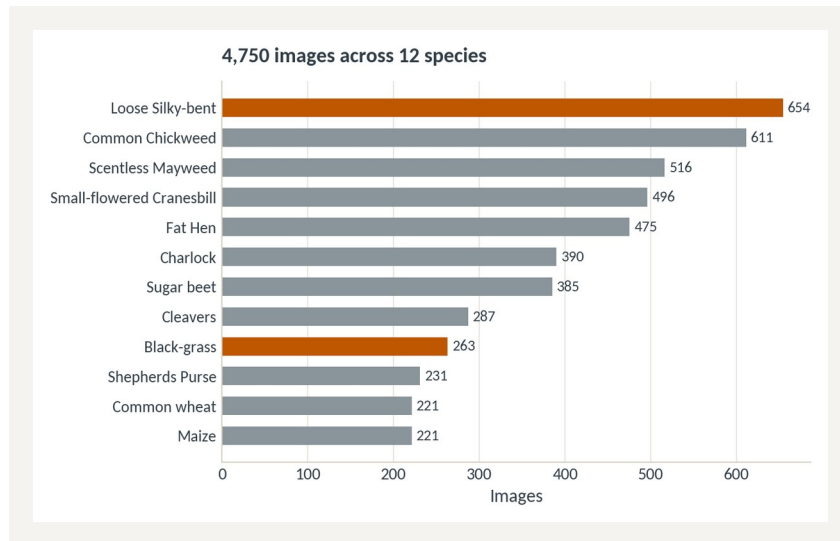
- Data preprocessing: color conversion, resizing, normalization, one-hot targets
- Baseline CNN
- Iterative development: augmentation, depth, batch normalization, transfer learning, L2, class weights
- Evaluation on a held-out test set; per-class precision, recall and confusion
- Final model selection and deployment recommendations

EDA results

The data

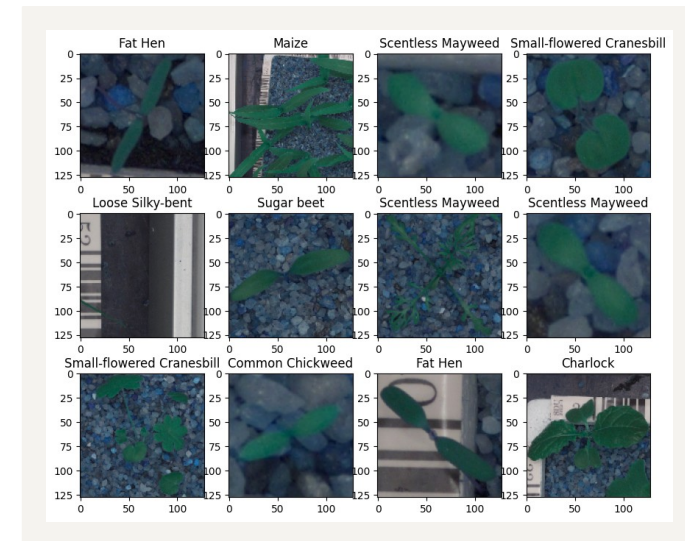
Twelve species, uneven representation

- Loose Silky-bent (654 images) has two and a half times as many examples as Black-grass (263) — and the two look alike. Common wheat and Maize are the smallest classes at 221.



What the model sees

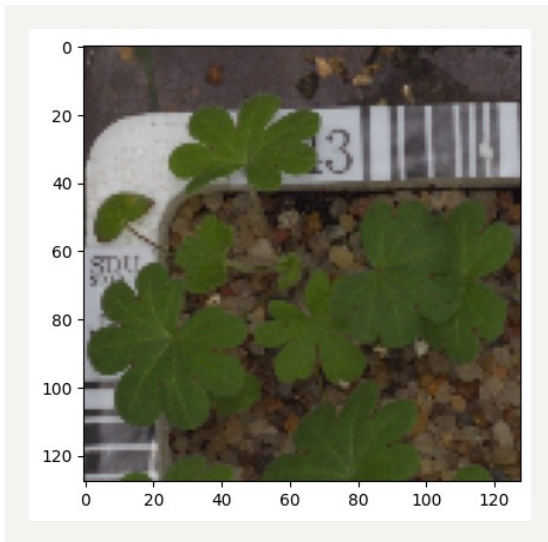
- Seedlings photographed against gravel; leaf shape, color and scale are the discriminating features.



Data preprocessing

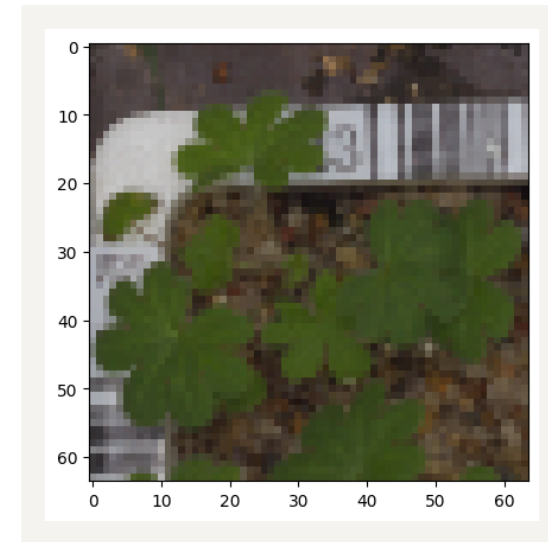
128 × 128

- Original resolution, BGR converted to RGB.



64 × 64

- Resized for training; pixel values scaled to 0–1 as float32. Splits: training 3,847 · validation 428 · test 475, targets one-hot encoded (12 classes).



Initial model

71% accuracy

Initial model — architecture

Sequential CNN • 128,828 parameters

Conv 128 • 3×3 • ReLU • same padding → max-pool 2×2

Conv 64 • 3×3 • ReLU • same → max-pool 2×2

Conv 32 • 3×3 • ReLU • same → max-pool 2×2

Flatten → **Dense 16 • ReLU** → **Dropout 0.3**

Output: Dense 12

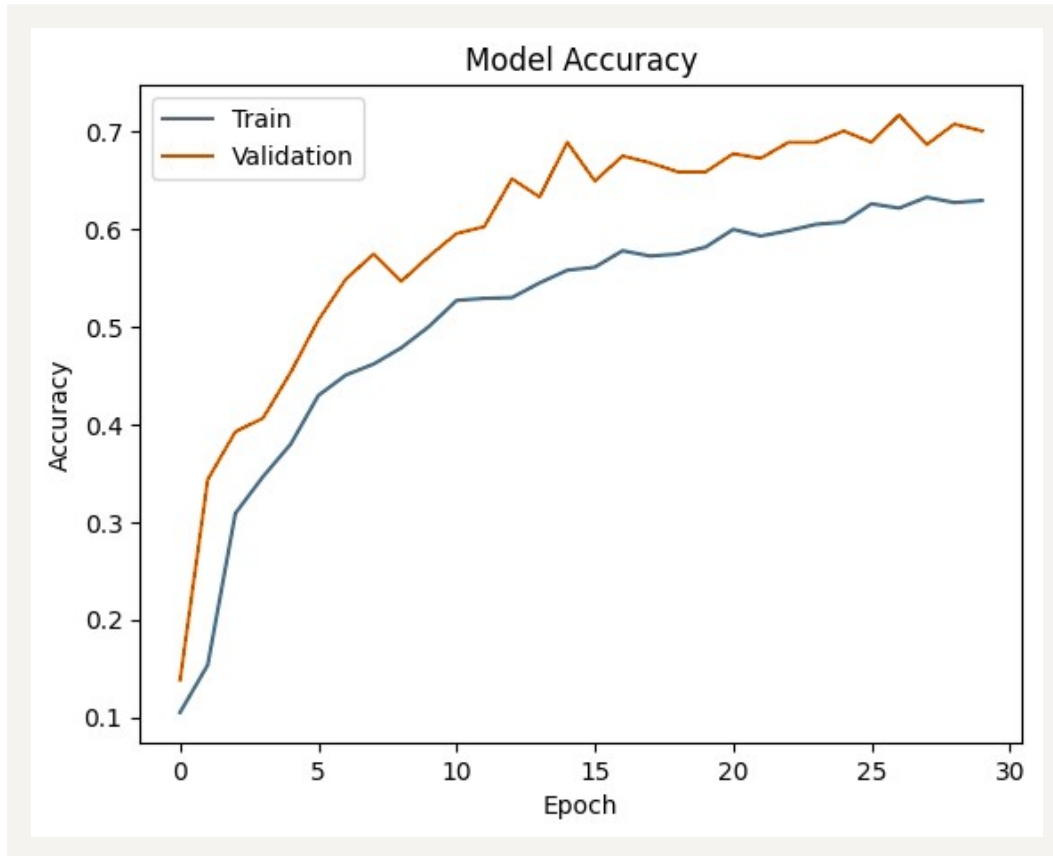
Training

Input: $64 \times 64 \times 3$.

Optimizer: Adam. Loss: categorical cross-entropy. Metric: accuracy.

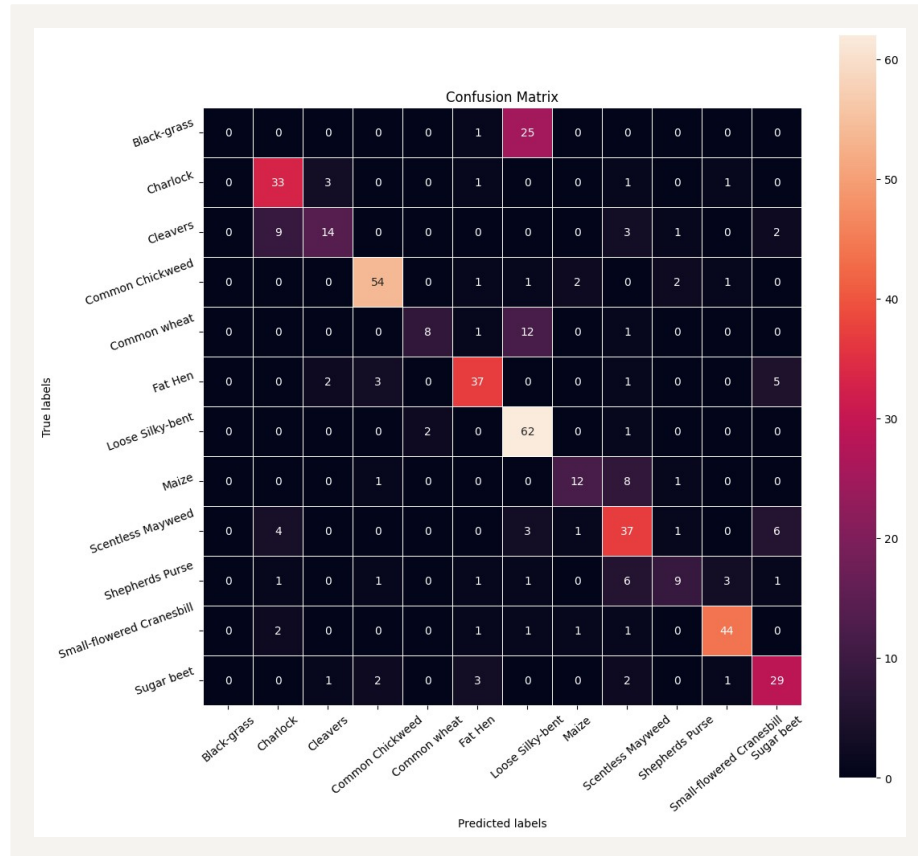
30 epochs; no augmentation, no regularization, no class weighting.

Initial model — training



- Training and validation accuracy rise quickly through the first ten epochs.
- Validation accuracy overtakes training around epoch 5 and stays above it — dropout depresses the training score, so the gap is not overfitting.
- Validation stabilizes near 70% and training near 63% by epoch 30.

Initial model — where it fails



Black-grass: zero recall. 25 of 26 test images predicted as Loose Silky-bent.

Common wheat: precision 0.80, recall 0.36 — rarely predicted.

Shepherds Purse: precision 0.64, recall 0.39; confused with Scentless Mayweed.

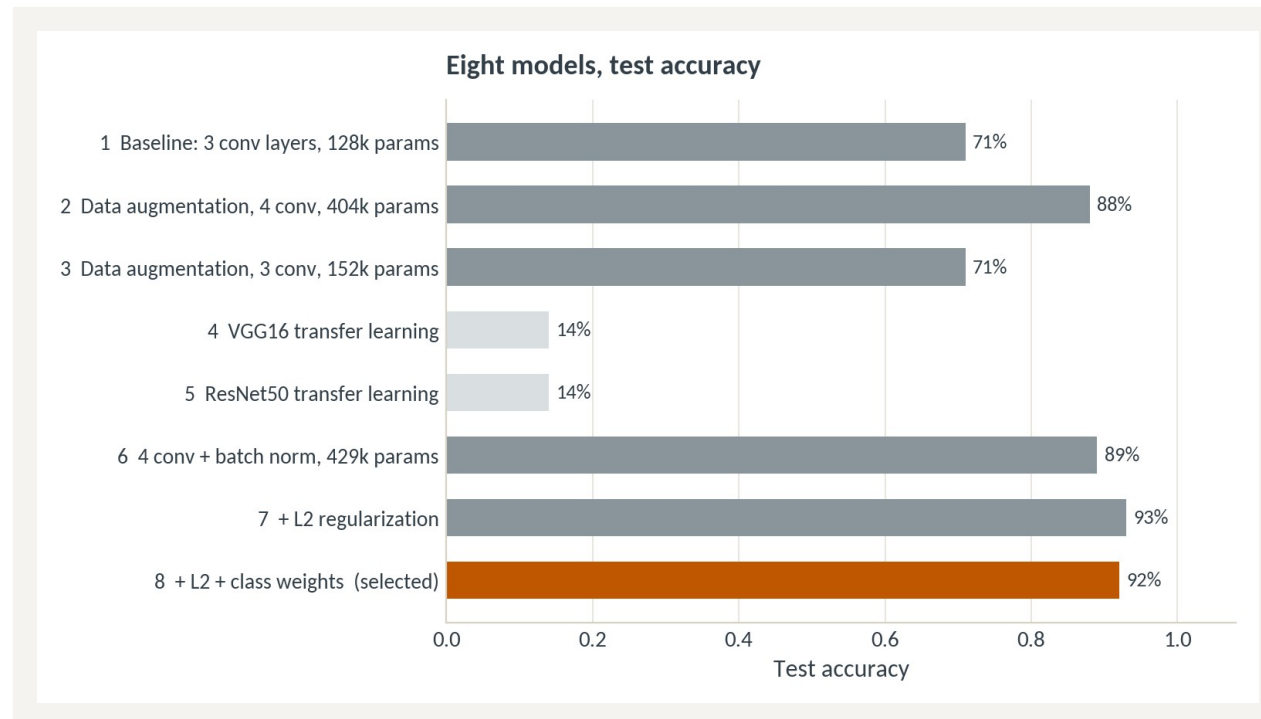
Common Chickweed is the one strong class: 54 of 61 correct.

Macro F1 0.63 against weighted 0.69: the model does better on the classes with more examples.

Model development

Eight architectures

Eight models, one ladder



Augmentation alone (model 2): horizontal flip, 20° rotation, nearest-fill — 88% with a wider four-layer network.

Transfer learning (4, 5): VGG16 and ResNet50 with frozen ImageNet bases collapsed to the majority class (14%). Small, low-resolution inputs and no fine-tuning.

Batch normalization (6): 89%.

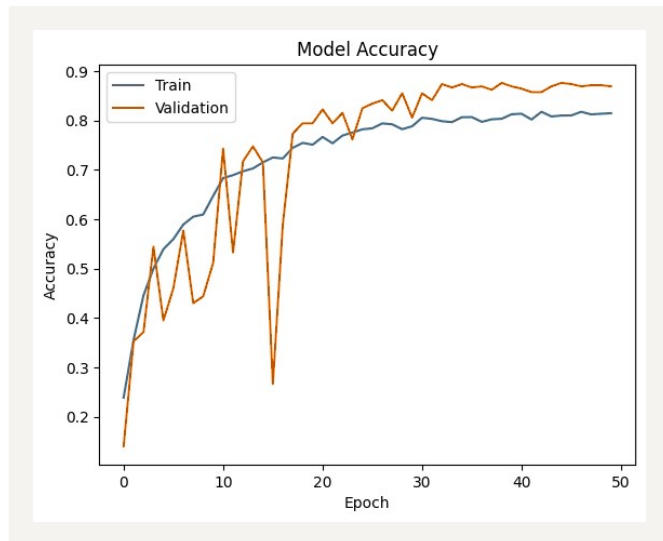
L2 regularization (7): 93% — the single largest gain after augmentation.

Class weights (8): 92% overall, chosen for more even recall across classes; the model taken forward.

Two instructive failures

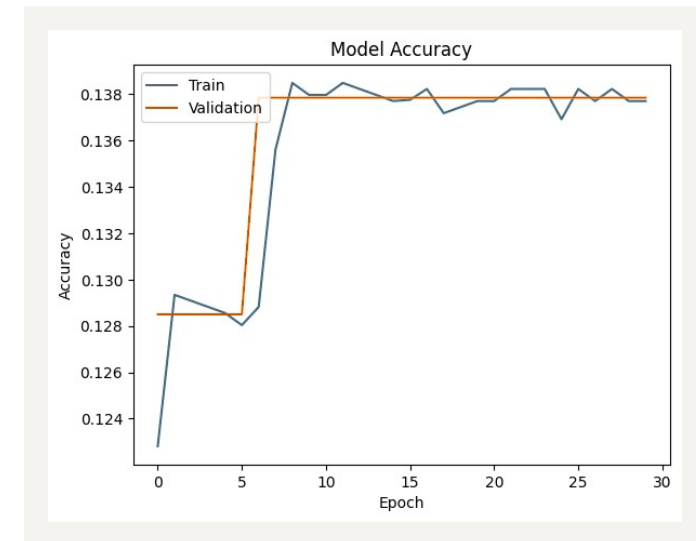
Model 2 — augmentation without regularization

- Validation accuracy swings by 40 points between epochs before settling; augmentation adds variance the network cannot yet absorb.



Model 4 — VGG16, frozen base

- Accuracy locks at 0.138 from epoch 7: the network outputs the majority class for every image.



Final model

92% accuracy

Final model — architecture

Sequential CNN • 428,972 parameters (428,524 trainable)

Conv 256 • 3×3 • ReLU • same → max-pool

Conv 128 • 3×3 • ReLU • L2 0.01 → max-pool → batch norm

Conv 64 • 3×3 • ReLU • L2 0.01 → max-pool → batch norm

Conv 32 • 3×3 • ReLU • L2 0.01 → max-pool → batch norm

Flatten → **Dense 64 • ReLU • L2** → **Dropout 0.3**

Output: Dense 12 • softmax • L2

Training strategy

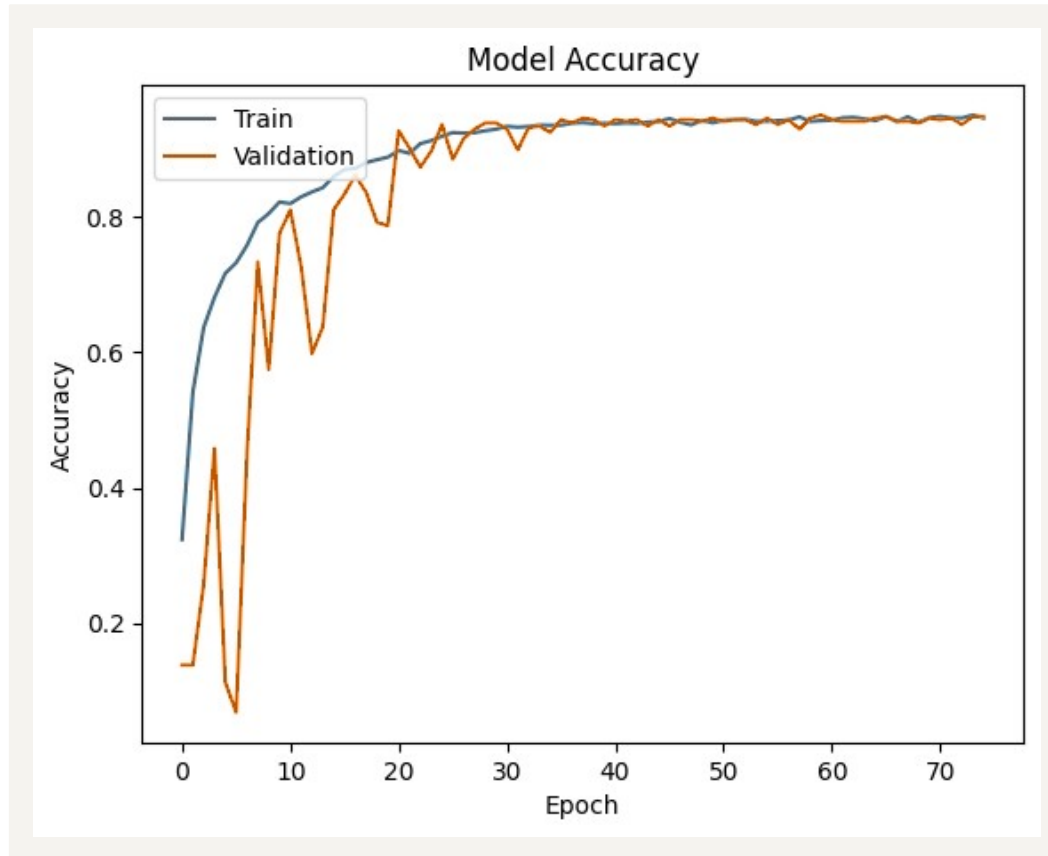
Augmentation: horizontal flip, rotation up to 20°, nearest-neighbor fill.

Balanced class weights computed from the training distribution.

ReduceLROnPlateau: monitor validation accuracy, patience 3, factor 0.5, floor 1e-5.

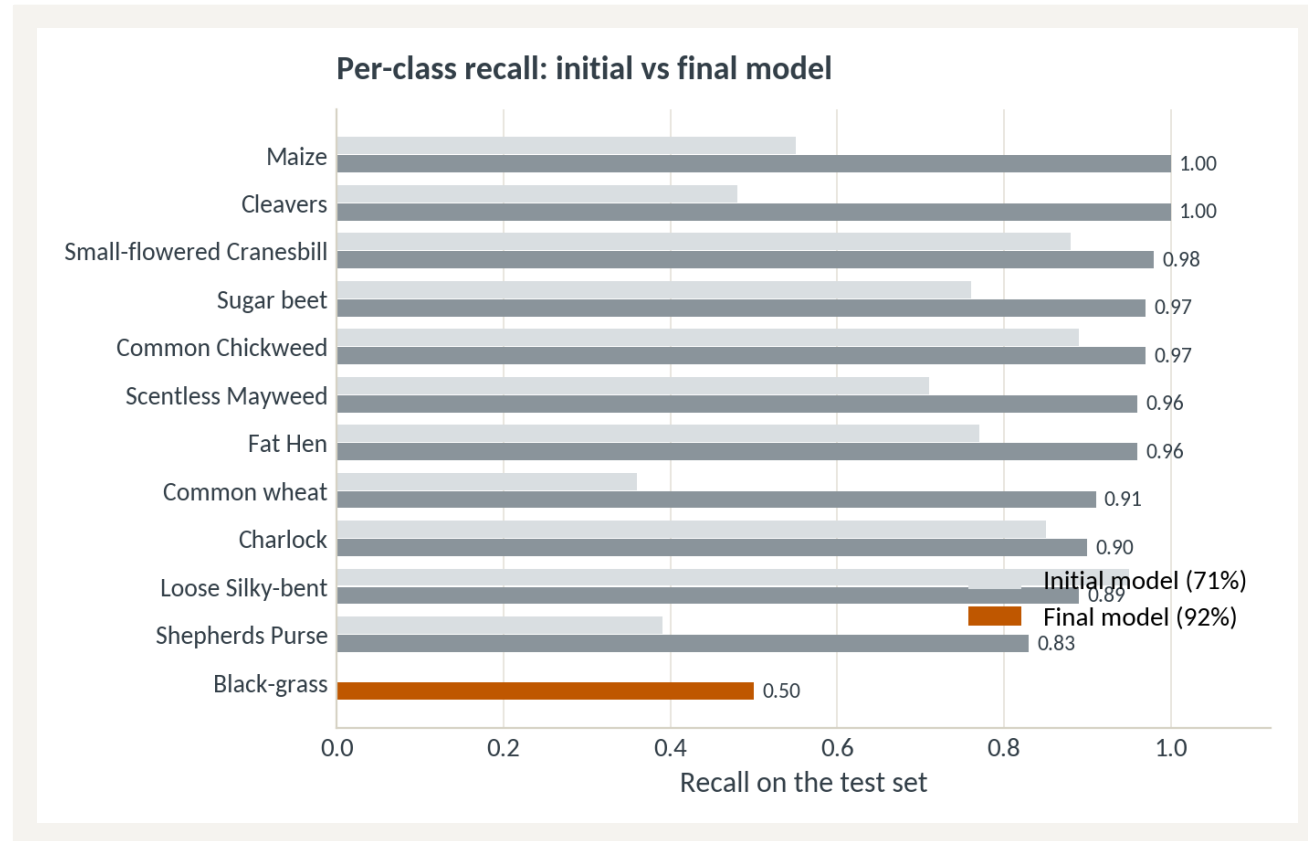
Adam; categorical cross-entropy; batch size 20.

Final model — training



- Training accuracy climbs steeply in the first ten epochs; validation oscillates early while the learning rate is high.
- Both curves rise together after epoch 20 and converge above 90%.
- The final training and validation accuracies sit within a point of each other — no sign of overfitting.

Final model — per-class performance



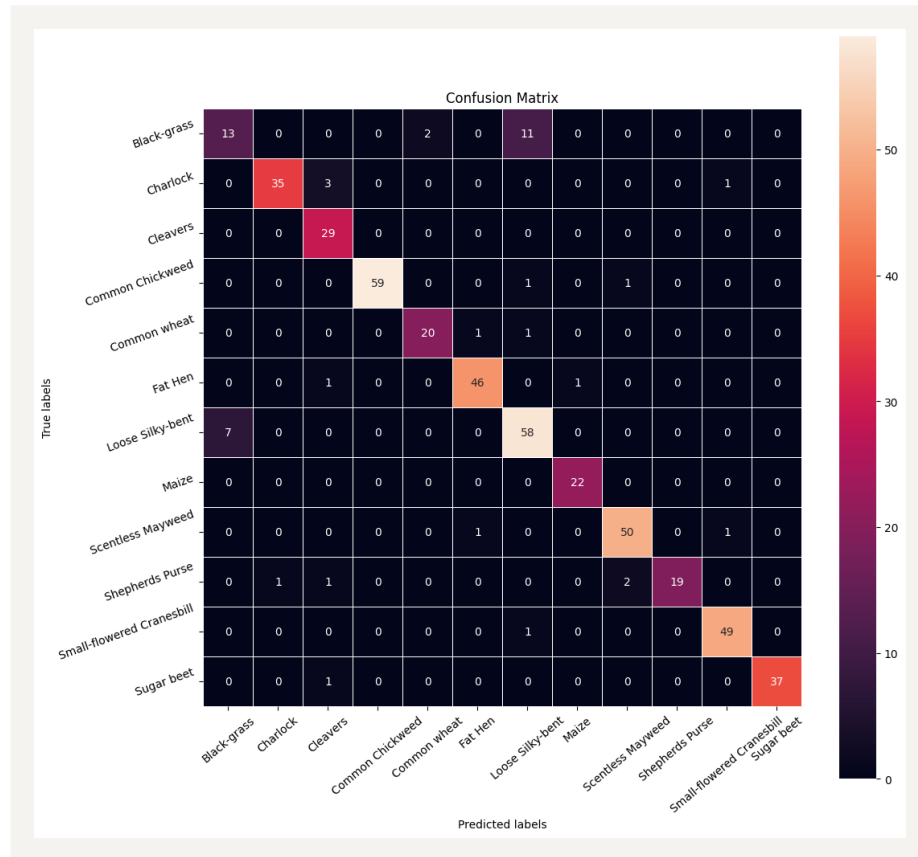
Nine of twelve classes at 0.90 recall or better; Maize and Cleavers at 1.00.

Black-grass remains the weak class: recall 0.50, precision 0.65, still confused with Loose Silky-bent (0.89).

Precision, recall and F1 sit close together for every class except Shepherds Purse (precision 1.00, recall 0.83).

Macro F1 0.91, weighted 0.92: the gap between rare and common classes has closed.

Final model — confusion



Black-grass ↔ Loose Silky-bent accounts for most remaining errors in both directions. The features that separate these grasses are not captured at 64×64 .

Errors elsewhere are scattered, one or two per class, with no second systematic confusion.

Class weights worked: misclassifications are spread rather than concentrated on the small classes.

Conclusion

The model is deployable for most species at 92% accuracy, with balanced precision and recall.

Black-grass and Loose Silky-bent need targeted work: more images across environments and conditions; texture or spectral features; color-shift and perspective augmentation aimed at the pair.

Next architectures: fine-tuned ResNet, or a hybrid of VGG and ResNet features, evaluated at native resolution.

Operationally: a retraining pipeline and a user feedback loop matter more than a further point of accuracy.



Thank you

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